

Lecture 6 – First-Order Functions

COSE212: Programming Languages

Jihyeok Park



2026 Fall

- VAE – AE **with variables**
 - Evaluation with Environments
 - Interpreter and Natural Semantics

- VAE – AE **with variables**
 - Evaluation with Environments
 - Interpreter and Natural Semantics

- In this lecture, we will learn **first-order functions**.

- VAE – AE **with variables**
 - Evaluation with Environments
 - Interpreter and Natural Semantics

- In this lecture, we will learn **first-order functions**.

- F1VAE – VAE **with first-order functions**
 - Concrete and Abstract Syntax
 - Interpreter and Natural Semantics
 - Static Scoping vs Dynamic Scoping

1. First-Order Functions

2. F1VAE – VAE with First-Order Functions

Concrete Syntax

Abstract Syntax

3. Interpreter and Natural Semantics for F1VAE

Evaluation with Function Environments

Function Application

Example

4. Static Scoping vs Dynamic Scoping

Example

1. First-Order Functions

2. F1VAE – VAE with First-Order Functions

Concrete Syntax

Abstract Syntax

3. Interpreter and Natural Semantics for F1VAE

Evaluation with Function Environments

Function Application

Example

4. Static Scoping vs Dynamic Scoping

Example

Let's calculate the square of several numbers in Scala.

```
3 * 3      // 9
42 * 42     // 1764
2434 * 2434 // 5924356
```

Let's calculate the square of several numbers in Scala.

```
3 * 3      // 9
42 * 42     // 1764
2434 * 2434 // 5924356
```

With a **first-order function**, we can avoid the repetition of the code.

```
// A `square` function that takes an integer `n` and returns its square.
def square(n: Int): Int = n * n

square(3)      // 9
square(42)     // 1764
square(2434)   // 5924356
```

A **first-order function** consists of 1) a function name, 2) a list of parameters, and 3) a function body, and it can be **invoked** with arguments.

Most programming languages support **first-order functions**.

- Scala

```
def square(n: Int): Int = n * n
```

```
square(3) // 9
```

- Python

```
def square(n): return n * n
```

```
square(3) # 9
```

- C++

```
int square(int n) { return n * n; }
```

```
square(3) // 9
```

- Rust

```
fn square(n: i32) -> i32 { return n * n; }
```

```
square(3) // 9
```

- ...

1. First-Order Functions

2. F1VAE – VAE with First-Order Functions

Concrete Syntax

Abstract Syntax

3. Interpreter and Natural Semantics for F1VAE

Evaluation with Function Environments

Function Application

Example

4. Static Scoping vs Dynamic Scoping

Example

Now, we want to extend VAE into F1VAE with **first-order functions**.

```
/* F1VAE */  
def square(n) = n * n;  
square(3) + 2 // 11
```

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

Now, we want to extend VAE into F1VAE with **first-order functions**.

```
/* F1VAE */  
def square(n) = n * n;  
square(3) + 2 // 11
```

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

- An F1VAE **program** is a pair of
 - 1 a list of **function definitions**
 - 2 an **expression**

Now, we want to extend VAE into F1VAE with **first-order functions**.

```
/* F1VAE */  
def square(n) = n * n;  
square(3) + 2 // 11
```

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

- An F1VAE **program** is a pair of
 - 1 a list of **function definitions**
 - 2 an **expression**
- A **function definition** in F1VAE only has a **single parameter**.

Now, we want to extend VAE into F1VAE with **first-order functions**.

```
/* F1VAE */  
def square(n) = n * n;  
square(3) + 2 // 11
```

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

- An F1VAE **program** is a pair of
 - 1 a list of **function definitions**
 - 2 an **expression**
- A **function definition** in F1VAE only has a **single parameter**.
- We extend **expressions** with **function applications**.

Let's define the **concrete syntax** of F1VAE in BNF:

- An F1VAE **program** is a pair of
 - ① a list of **function definitions**
 - ② an **expression**
- A **function definition** in F1VAE only has a **single parameter**.
- We extend **expressions** with **function applications**.

Let's define the **concrete syntax** of F1VAE in BNF:

- An F1VAE **program** is a pair of
 - ① a list of **function definitions**
 - ② an **expression**
- A **function definition** in F1VAE only has a **single parameter**.
- We extend **expressions** with **function applications**.

```
// programs
<program> ::= <fdef>* <expr>

// function definitions
<fdef>    ::= "def" <id> "(" <id> ")" "=" <expr> ";"

// expressions
<expr>    ::= ...
            | "{" <expr> "}"                // block
            | "val" <id> "=" <expr> ";" <expr> // variable definition
            | <id>                          // identifier lookup
            | <id> "(" <expr> ")"            // function application
```

Let's define the **abstract syntax** of F1VAE in BNF and its **ADT** in Scala:

Programs	$p ::= f^* e$	(Program)
Function Definitions	$\mathbb{F} \ni f ::= \text{def } x(x) = e$	(FunDef)
Expressions	$e ::=$	...
	$\text{val } x = e; e$	(Val)
	x	(Id)
	$x(e)$	(App)

Let's define the **abstract syntax** of F1VAE in BNF and its **ADT** in Scala:

Programs	$p ::= f^* e$	(Program)
Function Definitions	$\mathbb{F} \ni f ::= \text{def } x(x) = e$	(FunDef)
Expressions	$e ::=$	...
	$\text{val } x = e; e$	(Val)
	x	(Id)
	$x(e)$	(App)

```
// programs
case class Program(fdefs: List[FunDef], expr: Expr)
// function definitions
case class FunDef(name: String, param: String, body: Expr)
// expressions
enum Expr:
  ...
  case Val(name: String, init: Expr, body: Expr)
  case Id(name: String)
  case App(fname: String, arg: Expr) // function application
```

For example, let's **parse** the following F1VAE program:

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4))
```

Then, the following **abstract syntax tree (AST)** is produced:

```
Program(  
  List(  
    FunDef("add3", "n", Add(Id("n"), Num(3))),  
    FunDef("mul2", "m", Mul(Id("m"), Num(2)))  
  ),  
  App("mul2", App("add3", Num(4)))  
)
```

1. First-Order Functions

2. F1VAE – VAE with First-Order Functions

Concrete Syntax

Abstract Syntax

3. Interpreter and Natural Semantics for F1VAE

Evaluation with Function Environments

Function Application

Example

4. Static Scoping vs Dynamic Scoping

Example

Let's evaluate the following F1VAE program:

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

Let's evaluate the following F1VAE program:

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

How to find the function definition of `add3` or `mul2`?

Let's evaluate the following F1VAE program:

```
/* F1VAE */  
def add3(n) = n + 3;  
def mul2(m) = m * 2;  
mul2(add3(4)) // 14
```

How to find the function definition of `add3` or `mul2`?

We need to construct a **function environment** that maps function names to function definitions from the **list of function definitions** in a program.

$$[\text{add3} \mapsto f_0, \text{mul2} \mapsto f_1]$$

where

$$\begin{aligned} f_0 &= \text{def add3}(n) = n + 3 \\ f_1 &= \text{def mul2}(m) = m * 2 \end{aligned}$$

For VAE, the interpreter takes an **expression** e with an **environment** σ and returns a number n as the result.

```
type Value = BigInt                                // values
type Env = Map[String, Value]                      // environments
def interp(expr: Expr, env: Env): Value = ... // interpreter
```

$$\sigma \vdash e \Rightarrow n$$

For VAE, the interpreter takes an **expression** e with an **environment** σ and returns a number n as the result.

```
type Value = BigInt                                // values
type Env = Map[String, Value]                      // environments
def interp(expr: Expr, env: Env): Value = ... // interpreter
```

$$\sigma \vdash e \Rightarrow n$$

Now, we extend it to take a **function environment** Λ , a mapping from function names to function definitions, as well:

```
type Value = BigInt                                // values
type Env = Map[String, Value]                      // environments
type FEnv = Map[String, FunDef]                    // function environments
def interp(expr: Expr, env: Env, fenv: FEnv): Value = ... // interpreter
```

$$\sigma, \Lambda \vdash e \Rightarrow n$$

We have a **list of function definitions** in an F1VAE program, not a **function environment**, a mapping from function names to definitions.

```
case class Program(fdefs: List[FunDef], expr: Expr)
type FEnv = Map[String, FunDef]
```

How to construct a FEnv from a List[FunDef]?

We have a **list of function definitions** in an F1VAE program, not a **function environment**, a mapping from function names to definitions.

```
case class Program(fdefs: List[FunDef], expr: Expr)
type FEnv = Map[String, FunDef]
```

How to construct a FEnv from a List[FunDef]?

```
def createFEnv(fdefs: List[FunDef]): FEnv =
  fdefs.foldLeft(Map.empty)((fenv: FEnv, fdef: FunDef) => {
    val fname: String = fdef.name
    if (fenv.contains(fname)) error(s"duplicate function: $fname")
    else fenv + (fname -> fdef)
  })
```

We have a **list of function definitions** in an F1VAE program, not a **function environment**, a mapping from function names to definitions.

```
case class Program(fdefs: List[FunDef], expr: Expr)
type FEnv = Map[String, FunDef]
```

How to construct a FEnv from a List[FunDef]?

```
def createFEnv(fdefs: List[FunDef]): FEnv =
  fdefs.foldLeft(Map.empty)((fenv: FEnv, fdef: FunDef) => {
    val fname: String = fdef.name
    if (fenv.contains(fname)) error(s"duplicate function: $fname")
    else fenv + (fname -> fdef)
  })
```

It will throw an error if there are **duplicate function names**:

```
createFEnv(List(
  FunDef("add3", "n", Add(Id("n"), Num(3))),
  FunDef("add3", "n", Add(Num(3), Id("n"))),
)) // error: duplicate function: add3
```

For F1VAE, we need to 1) implement the **interpreter**:

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = ???
```

For F1VAE, we need to 1) implement the **interpreter**:

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = ???
```

and 2) define the **natural semantics** with environments and **function environments**:

$$\sigma, \Lambda \vdash e \Rightarrow n$$

Programs	$p ::= f^* e$	(Program)
Function Definitions	$\mathbb{F} \ni f ::= \text{def } x(x) = e$	(FunDef)
Expressions	$e ::= \dots$	
	$\quad \quad x(e)$	(App)

where

Numbers	$n \in \mathbb{Z}$	(BigInt)
Identifiers	$x \in \mathbb{X}$	(String)
Environments	$\sigma \in \mathbb{X} \xrightarrow{\text{fin}} \mathbb{Z}$	(Env)
Function Environments	$\Lambda \in \mathbb{X} \xrightarrow{\text{fin}} \mathbb{F}$	(FEnv)

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
  ...
  case App(f, e) => ???
```

$$\sigma, \Lambda \vdash e \Rightarrow n$$

$$\text{App} \frac{\text{???}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow \text{???}}$$

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
...
case App(f, e) =>
  val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
  ...
```

$$\sigma, \Lambda \vdash e \Rightarrow n$$

$$\text{App} \frac{\begin{array}{c} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \dots \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow ???}$$

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
...
case App(f, e) =>
  val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
  ... interp(e, env, fenv) ...
```

$$\boxed{\sigma, \Lambda \vdash e \Rightarrow n}$$

$$\text{App} \frac{\begin{array}{c} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad \dots \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow ???}$$

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
...
case App(f, e) =>
  val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
  ... Map(fdef.param -> interp(e, env, fenv)) ...
```

$$\boxed{\sigma, \Lambda \vdash e \Rightarrow n}$$

$$\text{App} \frac{\begin{array}{c} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad \dots [x_1 \mapsto n_1] \dots \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow n_2}$$

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
...
case App(f, e) =>
  val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
  interp(fdef.body, Map(fdef.param -> interp(e, env, fenv)), fenv)
```

$$\boxed{\sigma, \Lambda \vdash e \Rightarrow n}$$

$$\text{App} \frac{\begin{array}{l} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad [x_1 \mapsto n_1], \Lambda \vdash e_2 \Rightarrow n_2 \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow n_2}$$

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
...
case App(f, e) =>
  val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
  interp(fdef.body, Map(fdef.param -> interp(e, env, fenv)), fenv)
```

$$\boxed{\sigma, \Lambda \vdash e \Rightarrow n}$$

$$\text{App} \frac{\begin{array}{l} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad [x_1 \mapsto n_1], \Lambda \vdash e_2 \Rightarrow n_2 \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow n_2}$$

We skip the other cases because they are only augmented with passing function environments. If you are interested, please refer to this spec:

<https://github.com/ku-plrg-classroom/docs/blob/main/cose212/f1vae/f1vae-spec.pdf>

```
/* F1VAE */
def inc(x) = x + 1;
inc(2) // 3
```

Let's evaluate the above F1VAE program:

$$\text{APP} \frac{\Lambda(\text{inc}) = f \quad \text{NUM} \frac{}{\emptyset, \Lambda \vdash 2 \Rightarrow 2} \quad \text{ADD} \frac{\text{ID} \frac{x \in \text{dom}(\sigma)}{\sigma, \Lambda \vdash x \Rightarrow 2} \quad \text{NUM} \frac{}{\sigma, \Lambda \vdash 1 \Rightarrow 1}}{\sigma, \Lambda \vdash x + 1 \Rightarrow 3}}{\emptyset, \Lambda \vdash \text{inc}(2) \Rightarrow 3}$$

where

$$\begin{aligned}
 \Lambda &= [\text{inc} \mapsto f] \\
 f &= \text{def inc}(x) = x + 1 \\
 \sigma &= [x \mapsto 2]
 \end{aligned}$$

1. First-Order Functions

2. F1VAE – VAE with First-Order Functions

Concrete Syntax

Abstract Syntax

3. Interpreter and Natural Semantics for F1VAE

Evaluation with Function Environments

Function Application

Example

4. Static Scoping vs Dynamic Scoping

Example

The current semantics is called **static scoping** (or **lexical scoping**) because the binding occurrence of an identifier is determined **statically** by where the function is **defined**, not where it is **applied**.

```
/* F1VAE */  
def f(x) = x + y;           // y is a free variable  
{ val y = 2; f(1) } + { val y = 4; f(3) }
```

The current semantics is called **static scoping** (or **lexical scoping**) because the binding occurrence of an identifier is determined **statically** by where the function is **defined**, not where it is **applied**.

```
/* F1VAE */  
def f(x) = x + y;           // y is a free variable  
{ val y = 2; f(1) } + { val y = 4; f(3) }
```

However, we can define the semantics of F1VAE in another way by using **dynamic scoping** instead; the binding occurrence is determined **dynamically** by the environment at the **call site** where the function is applied:

```
/* F1VAE */  
def f(x) = x + y;           // y = 2 or y = 4 depending on the call-site  
{ val y = 2; f(1) } + { val y = 4; f(3) } // (1 + 2) + (3 + 4) = 10
```

We can design and implement the semantics of F1VAE with **dynamic scoping** by changing the definition of the function application:

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
  ...
  case App(f, e) =>
    val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
    interp(fdef.body, env + (fdef.param -> interp(e, env, fenv)), fenv)
```

$$\sigma, \Lambda \vdash e \Rightarrow n$$

$$\text{App} \frac{\begin{array}{l} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad \sigma[x_1 \mapsto n_1], \Lambda \vdash e_2 \Rightarrow n_2 \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow n_2}$$

We can design and implement the semantics of F1VAE with **dynamic scoping** by changing the definition of the function application:

```
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
  ...
  case App(f, e) =>
    val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
    interp(fdef.body, env + (fdef.param -> interp(e, env, fenv)), fenv)
```

$$\sigma, \Lambda \vdash e \Rightarrow n$$

$$\text{App} \frac{\begin{array}{c} x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \\ \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad \sigma[x_1 \mapsto n_1], \Lambda \vdash e_2 \Rightarrow n_2 \end{array}}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow n_2}$$

However, we will use **static scoping** by default in this course.

```
/* F1VAE */
def add(x) = x + y;
val y = 2; add(1)
```

If we evaluate the above F1VAE program with **static scoping**:

$$\begin{array}{c}
 \text{NUM} \frac{}{\emptyset, \Lambda \vdash 2 \Rightarrow 2} \quad \text{APP} \frac{\Lambda(\text{add}) = f \quad \text{NUM} \frac{}{\sigma_0, \Lambda \vdash 1 \Rightarrow 1} \quad \text{ADD} \frac{\text{ID} \frac{x \in \text{dom}(\sigma_1)}{\sigma_1, \Lambda \vdash x \Rightarrow 1} \quad \text{ID} \frac{y \notin \text{dom}(\sigma_1)}{\sigma_1, \Lambda \vdash y \Rightarrow \text{FAIL}}}{\sigma_1, \Lambda \vdash x + y \Rightarrow \text{FAIL}}}{\sigma_0, \Lambda \vdash \text{add}(1) \Rightarrow \text{FAIL}} \\
 \text{VAL} \frac{}{\emptyset, \Lambda \vdash \text{val } y = 2; \text{ add}(1) \Rightarrow \text{FAIL}}
 \end{array}$$

where

$$\begin{aligned}
 \Lambda &= [\text{add} \mapsto f] \\
 f &= \text{def add}(x) = x + y \\
 \sigma_0 &= [y \mapsto 2] \\
 \sigma_1 &= [x \mapsto 1]
 \end{aligned}$$

it cannot be evaluated because y is a **free variable** in the add function.

```
/* F1VAE */
def add(x) = x + y;
val y = 2; add(1)
```

However, if we evaluate it with **dynamic scoping**:

$$\begin{array}{c}
 \text{VAL} \frac{\text{NUM} \frac{\text{APP} \frac{\text{ID} \frac{x \in \text{dom}(\sigma_1)}{\sigma_1, \Lambda \vdash x \Rightarrow 1} \quad \text{ID} \frac{y \in \text{dom}(\sigma_1)}{\sigma_1, \Lambda \vdash y \Rightarrow 2}}{\sigma_1, \Lambda \vdash x + y \Rightarrow 3}}{\sigma_0, \Lambda \vdash 1 \Rightarrow 1} \quad \Lambda(\text{add}) = f}{\sigma_0, \Lambda \vdash \text{add}(1) \Rightarrow 3}}{\emptyset, \Lambda \vdash 2 \Rightarrow 2} \\
 \hline
 \emptyset, \Lambda \vdash \text{val } y = 2; \text{ add}(1) \Rightarrow 3
 \end{array}$$

where

$$\begin{aligned}
 \Lambda &= [\text{add} \mapsto f] \\
 f &= \text{def add}(x) = x + y \\
 \sigma_0 &= [y \mapsto 2] \\
 \sigma_1 &= \sigma_0[x \mapsto 1] = [y \mapsto 2, x \mapsto 1]
 \end{aligned}$$

it evaluates to 3 because y is **dynamically bound** to 2.

Programs	$p ::= f^* e$	(Program)
Function Definitions	$\mathbb{F} \ni f ::= \text{def } x(x) = e$	(FunDef)
Expressions	$e ::= \dots$ $x(e)$	(App)

```

type FEnv = Map[String, FunDef]
def interp(expr: Expr, env: Env, fenv: FEnv): Value = expr match
  ...
  case App(f, e) =>
    val fdef = fenv.getOrElse(f, error(s"unknown function: $f"))
    interp(fdef.body, Map(fdef.param -> interp(e, env, fenv)), fenv)

```

$$\boxed{\sigma, \Lambda \vdash e \Rightarrow n}$$

$$\text{App} \frac{x_0 \in \text{dom}(\Lambda) \quad \Lambda(x_0) = \text{def } x_0(x_1) = e_2 \quad \sigma, \Lambda \vdash e_1 \Rightarrow n_1 \quad [x_1 \mapsto n_1], \Lambda \vdash e_2 \Rightarrow n_2}{\sigma, \Lambda \vdash x_0(e_1) \Rightarrow n_2}$$

<https://github.com/ku-plrg-classroom/docs/tree/main/cose212/f1vae>

- Please see above document on GitHub:
 - Implement `interp` function.
 - Implement `interpDS` function.
- It is just an exercise, and you **don't need to submit** anything.
- However, some exam questions might be related to this exercise, so we **strongly recommend** solving it by yourself.

- First-Class Functions

Jihyeok Park

`jihyeok_park@korea.ac.kr`

`https://plrg.korea.ac.kr`